Introduction to Machine Learning

Module 1: Chapter 1

Need for Machine Learning

- Business organization use huge amount of data for their daily activities.
- This data was not utilized due to two reasons:
  - **Data being scattered across** different archive systems and not able to integrate from these sources.
  - **Lack of awareness** about the software tools that could help to extract information from the data.
- ML is being adopted for this purpose.
- The 3 reasons for ML to become popular are:
  - **High volume of available data to manage:** The data being doubled every year, many big companies like Facebook, Twitter, YouTube generate huge of data.
  - **Cost of Storage:** Due to reduce in the storage cost, it is easier to capture, process, store, distribute and transmit the digital information.
  - **Availability of Complex algorithms:** These algorithms aids in the ML.

Terminologies in ML

- Data: All facts are data. It can be numbers or text that can be processed by the computer. Data sources can be flat files, databases or data warehouses in different storage formats.
- Information: Processed data is called information. This includes patterns, associations, or relationships among data. Eg: Sales data analyzed to extract information regarding the fast selling product.
- Knowledge: Condensed information is called knowledge. Unless the knowledge is extracted data is of no use. Eg: Historical patterns and future trends obtained in the above sales data can be called knowledge.
- Intelligence: Is applied knowledge for actions. An actionable form of knowledge is called intelligence.
- Wisdom: It is the ultimate objective of knowledge. It represents the maturity of mind. As of now exhibited by only humans.

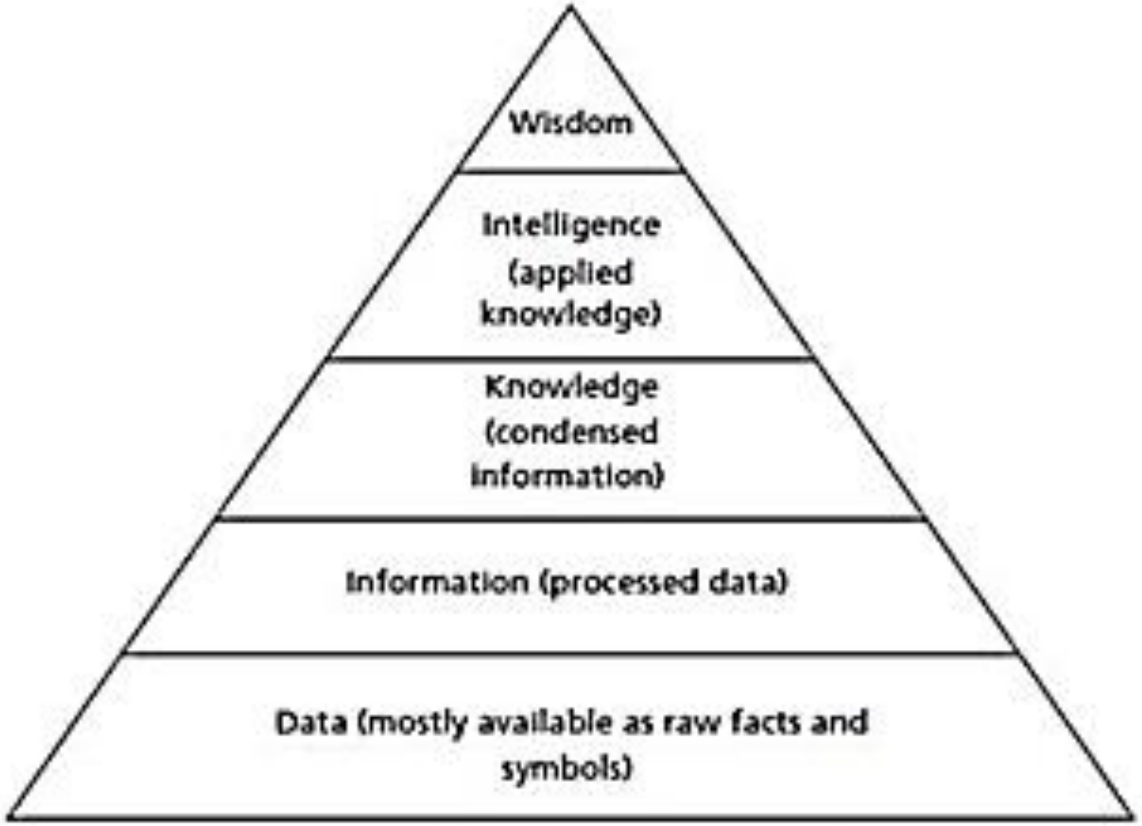


Figure 1.1: The Knowledge Pyramid

Machine Learning

- It is the sub-branch of Artificial Intelligence (AI).
- Arthur Samuel stated “Machine Learning is the field of study that gives the computers ability to learn without being explicitly programmed.
- Traditional method in programming is understanding the problem, a detailed design of the program in form flowchart or algorithm and the convert into programs using specific programming language. This method is not suitable for real-world problems like puzzle, games, complex image recognition etc...

Machine Learning

- In the beginning AI aimed at understanding the problems and develop the general rules manually. These rules are formulates into logic and implemented in a program to create intelligent systems. This idea of developing intelligent systems by using logic and reasoning by converting the expert's knowledge into a set of rules and programs is called expert system. Eg: MYCIN is an example for expert system.
- The drawback of the above system is it still depended on the human expertise and hence do not truly exhibit intelligence.
- The focus of AI is to develop intelligent systems by using data-driven approach, where data is used as an input to develop intelligent models. The models can then be used to predict new inputs.
- The aim of ML is to learn a model or set of rules from the given dataset automatically so that it can predict the unknown data correctly.

Machine Learning

- Humans take decisions based on an experience, computers make models based on extracted patterns in the input data and then use these data-filled models for prediction and to take decisions.
- In Statistical learning, the relationship between the input x and output y is modeled as a function in the form y=f(x).
- f -learning function that maps the input x to output y. Learning of function is the crucial aspect of forming a model in statistical learning.
- The Learning program summarizes the raw data in a model. A model is an explicit description of patterns within the data in the form of
  - Mathematical Equation
  - Relational diagrams like tree/graphs
  - Logical if/else rules
  - Groupings called clusters
- Model can be a formula, procedure or representation that can generate data decisions.

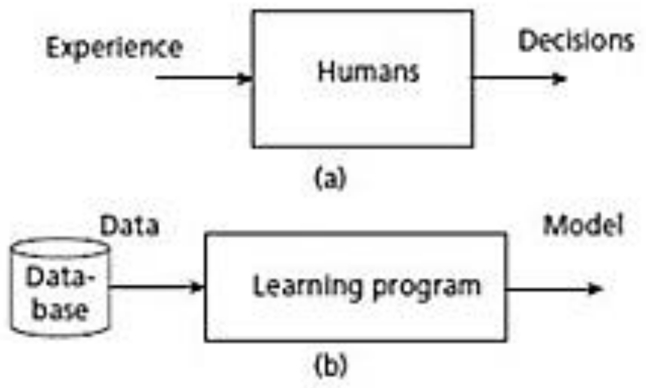


Figure 1.2: (a) A Learning System for Humans (b) A Learning System for Machine Learning

Machine Learning

- Difference between pattern and model
  - Pattern is local and applicable only to certain attributes.
  - Model is global and fits the entire dataset.
- Tom Mitchell's definition of ML: " A computer program is said to learn from experience E, with respect to task T and some performance measure P, if its performance on T measured by P improves with experience E". The key terms in this definition are Experience E, task T and performance measure P.
- Eg: Let task T is to detect an object in an image. The machine can gain the knowledge of the object using training dataset of thousands of images. This is called experience E.

Machine Learning

- Using this experience E for the task of object detection T, and is able to detect the object is measured by performance measures like precision and recall.
- Computers gather experience by the following steps
  - Collection of data.
  - Once the data is gathered, abstract concepts are formed out of that data. Abstraction is used to generate concepts.
  - Generalization converts the abstraction into an actionable form of intelligence. Generalization involves ranking of concepts, inferencing from them and formation of heuristics, an actionable aspect of intelligence. Heuristics are educated guesses for all tasks.
  - Usually heuristics normally works, sometimes it may fail. Hence some course of correction is done by taking evaluation measures.

Machine Learning in relation to other fields

- Machine Learning and AI:
- ML is an important branch of AI whose aim is to extract the patterns for prediction.
- Aim of AI is to develop intelligent agents.
- An agent can be robot, humans or any autonomous systems.
- AI winters – Idea of AI is to develop Intelligent agents and focus was on logic and logical inferences. The ups and downs of AI in building the Intelligent agents is called AI winters.
- It is a broad field that includes learning from examples and other areas like reinforcement learning.
- Deep Learning is a subbranch of ML using the neural network technology.

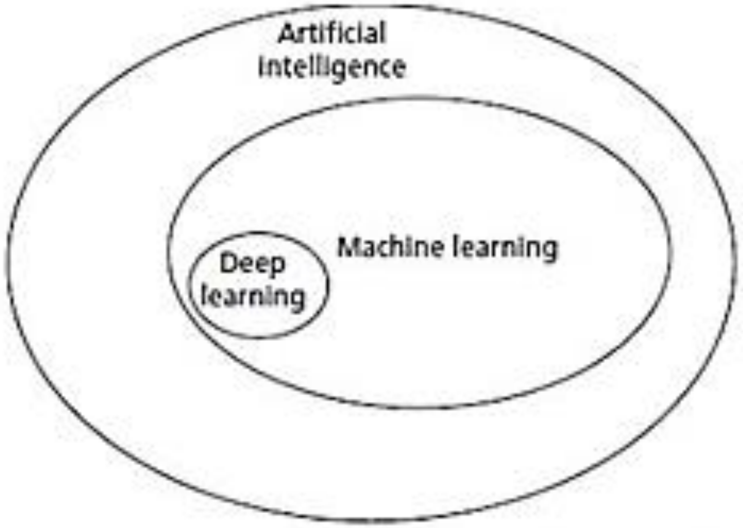


Figure 1.3: Relationship of AI with Machine Learning

Machine Learning, Data Science, Data Mining, and Data Analytics

- Data Science: Is an umbrella term that encompasses many fields.
- ML is a branch of Data science and starts with data.
- Data science deals with gathering of data for analysis. It is broad field that includes:
- Big data: It is a field of data science that deals with data's characteristics :
    - Volume: Huge amount of data is generated by big companies like twitter, Facebook
    - Variety: Data is available in variety of formats like images, videos, text…
    - Velocity: refers to the speed at which the data is generated and processed.
- Big data is used by ML for applications like language translation and image recognition. Big data influences the growth of Deep Learning.

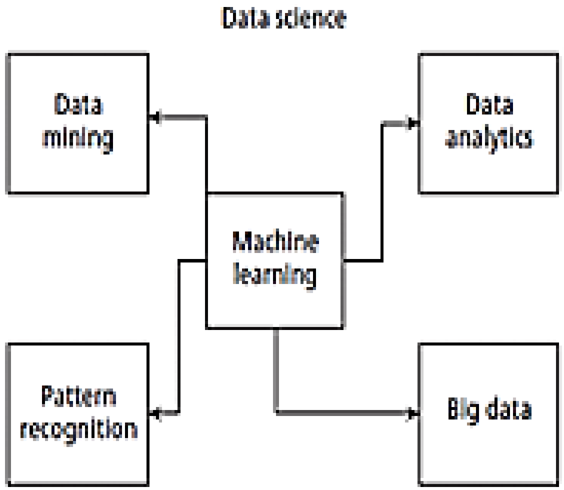


Figure 1.4: Relationship of Machine Learning with Other Major Fields

Machine Learning and Statistics:

- Statistics is a branch of mathematics that has theoretical foundation regarding statistical learning.
- Statistical methods look for regularity in data called patterns.
- Statistics sets a hypothesis and performs experiments to verify and validate the hypothesis in order to find the relationships among data.
- Statistics require knowledge of the statistical procedures and the guidance of a good statistician.
- It is mathematics intensive and models are often complicated equations and involve many assumptions. They are developed in relation to the data being analyzed.
- They are coherent and rigorous. It has strong theoretical foundations and interpretations that require a strong statistical knowledge.
- ML has less assumptions and requires less statistical knowledge. It interacts with various tools to automate the process of learning.

Types of Machine Learning:

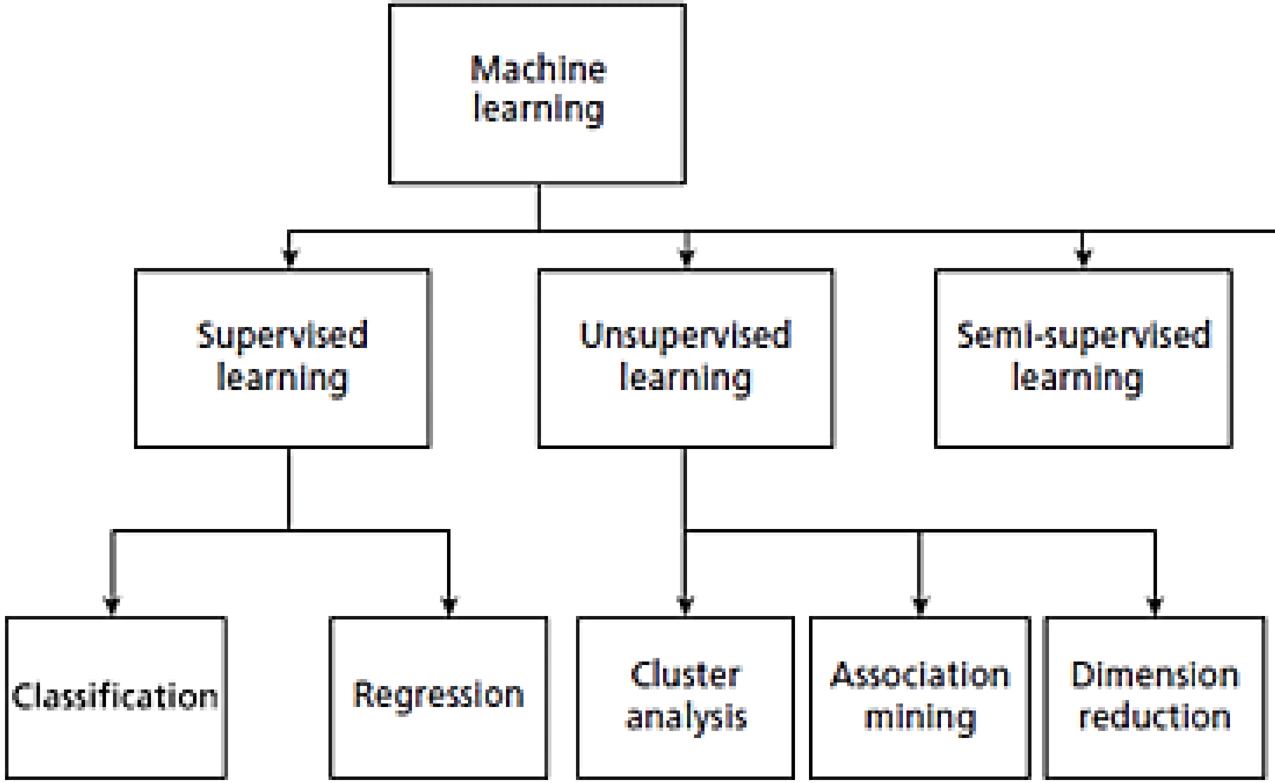


Figure 1.5: Types of Machine Learning

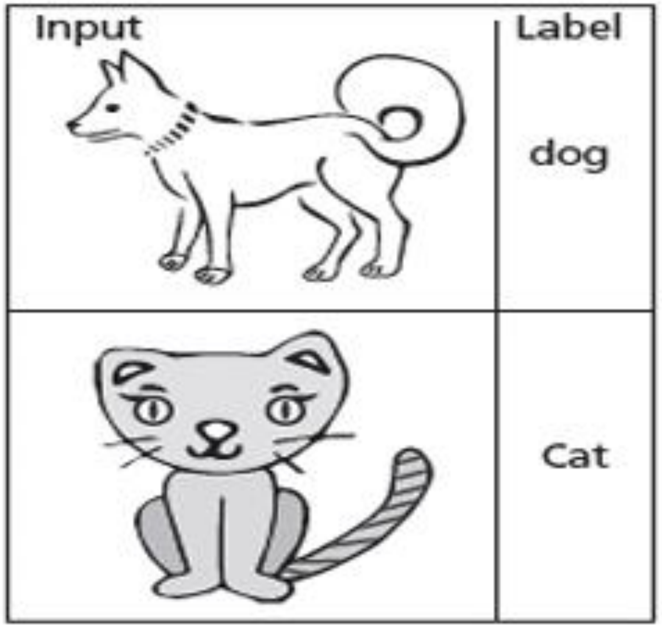
Labelled and Unlabelled data:

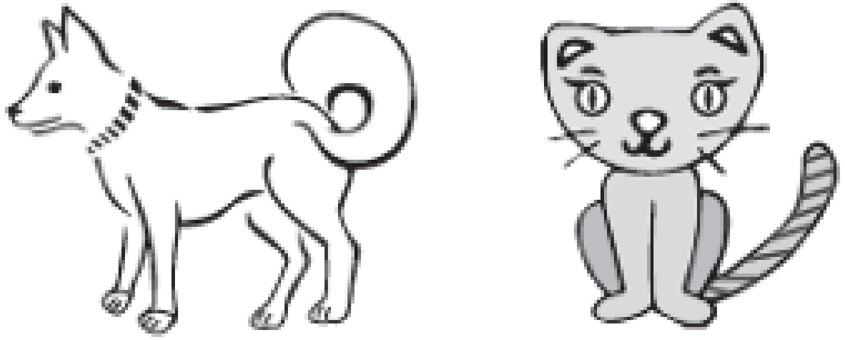
- Data is a raw fact. It is represented in the form of a table. Data can be referred as data point, sample or example. Each row of the table represent a data point.
- Features are attributes or characteristics of an object. Columns of the table are attributes. Out of all attributes, one attribute is called label. Label is the feature that we aim to predict.
- There are two types of data – labelled and unlabeled.
  - Labelled data - Labelled data is data that’s subject to a prior understanding of the way in which the world operates. A human or automatic tagger must use their prior knowledge to impose additional information on the data. Labelled data is carefully annotated with meaningful tags, or labels, that classify the data's elements or outcomes.
  - Unlabeled data: consists of raw inputs with no designated outcome.

Labelled and Unlabelled data:

Table 1.1: Iris Flower Dataset

S.No.	Length of Petal	Width of Petal	Length of Sepal	Width of Sepal	Class
1.	5.5	4.2	1.4	0.2	Setosa
2.	7	3.2	4.7	1.4	Versicolor
3.	7.3	2.9	6.3	1.8	Virginica





Supervised Learning

- Use **labelled** dataset.
- There is a **supervisor or teacher** component.
- A supervisor provides labelled data so that the model is constructed and generates test data.
- In supervised learning algorithms, learning takes place in 2 stages.
  - In the first stage the supervisor communicates the information to machine that the machine is supposed to master. The machine understands it.
  - In the second stage, the supervisor asks the machine with set of questions to find out whether the machine has grasped the information. Based on these questions the machine is assessed whether it has learnt or not.

Supervised Learning

- Supervised learning has 2 methods: **Classification and regression.**
- **Classification:**
- The input attributes of the classification algorithms are called **independent variables.**
- The target is called **label or dependent variable.**
- The relationship between the input and target variable is represented in the form of a structure called classification model.
- The focus of classification is to predict the label is in discrete form.
- In the figure below the classification algorithm takes a set of labelled data images such as dogs and cats to construct a model that can be used to classify an unknown test image data.

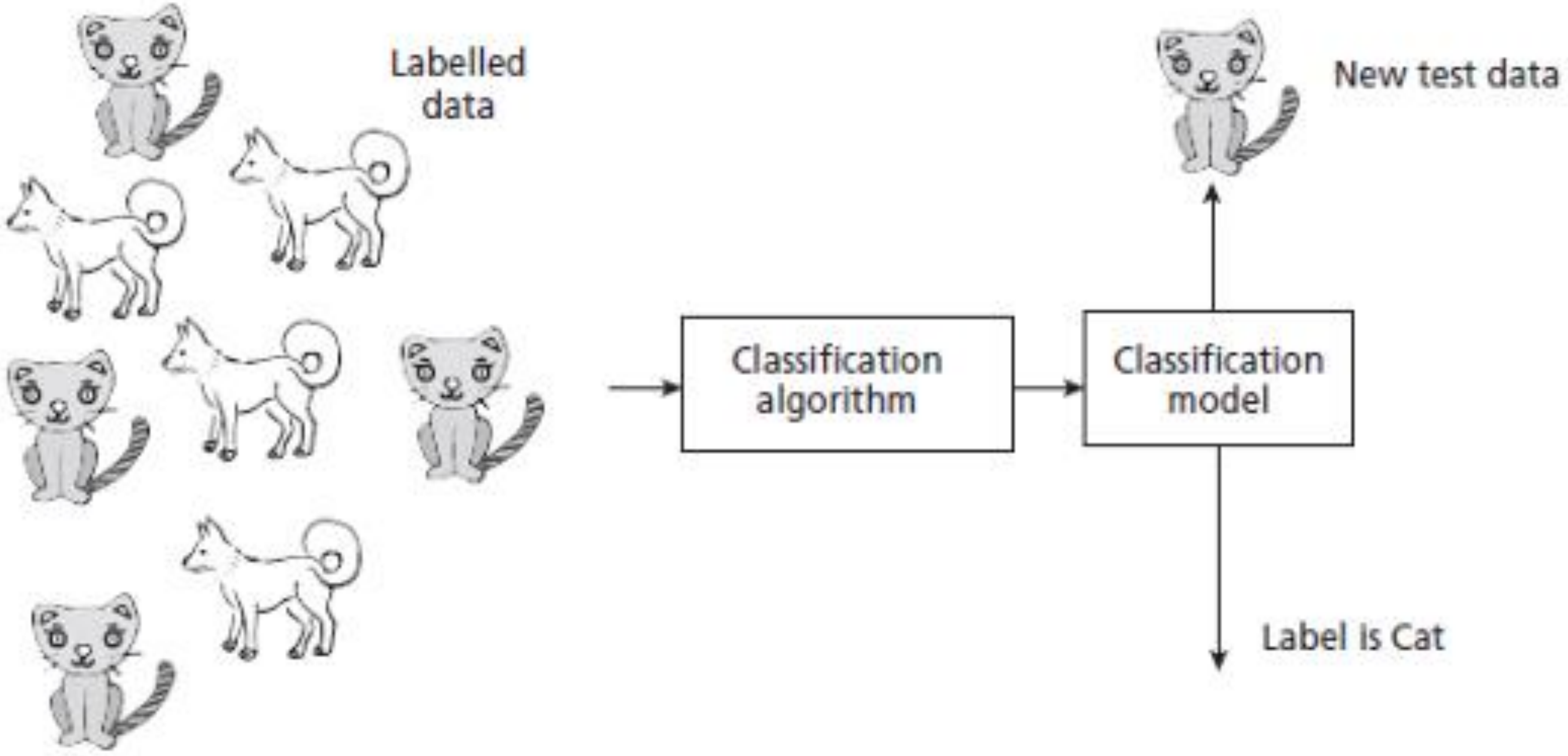


Figure 1.7: An Example Classification System

Supervised Learning

- In classification, learning takes place in two stages:
- First stage – called training stage, the learning algorithm takes a labelled dataset and starts learning. After the training set, samples are processed and the model is generated.
- Second stage – Constructed model is tested with test or unknown sample and assigned a label.
- Some of the key algorithms of classification are: Decision tree, Random forest, Support Vector machines, Naïve Bayes, Artifical Neural Network and Deep Learning Networks like CNN.

Supervised Learning

- Regression Models: It predicts the continuous variables like price. The regression model takes input x and generates a model in the form of a fitted line of the form y=f(x). X is independent variable and may be one or more attributes. Y is the dependent variable.

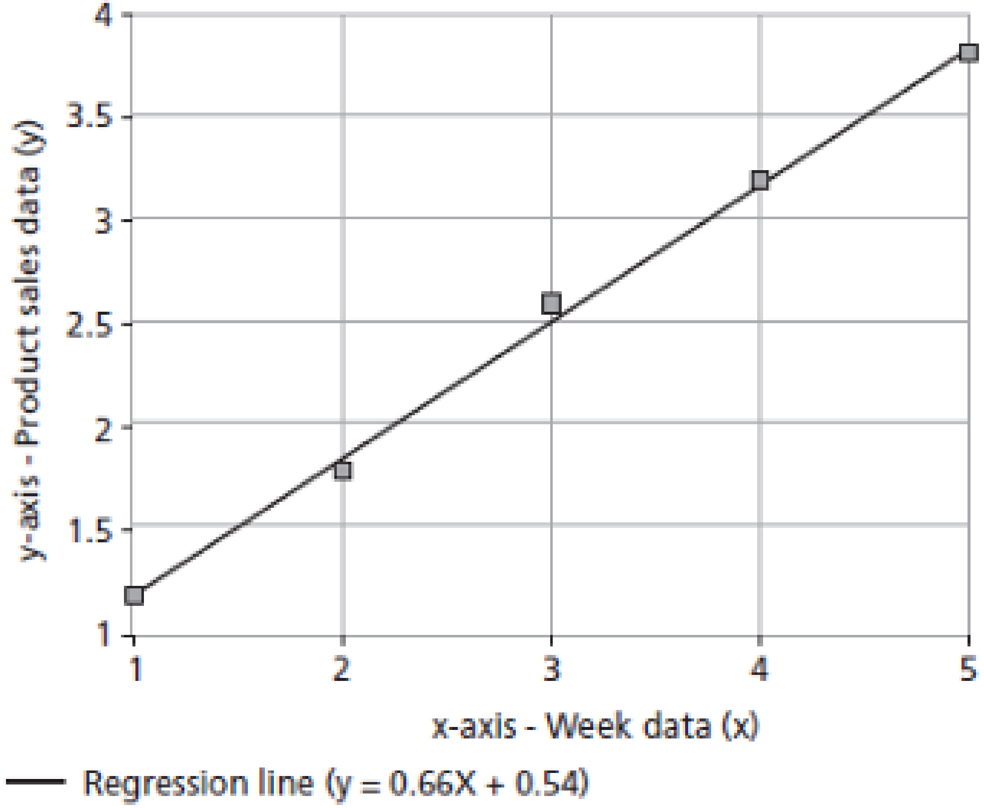


Figure 1.8: A Regression Model of the Form $y = ax + b$

Unsupervised Learning:

- Learning by self-instruction.
- No supervisor component.
- It is based on the concept of trial and error.
- The program is supplied with objects, but no labels are defined. The algorithm itself observes the examples and recognizes patterns based on the principles of grouping. Grouping is done in ways that similar objects form a same group.
- Cluster analysis and Dimensional reduction algorithms are examples of unsupervised algorithms.
- Cluster analysis aim to group objects into disjoint clusters or groups based on its attributes. Key clustering algorithms are k-means and Hierarchical algorithms.

Unsupervised Learning

- In the figure the clustering algorithms takes set of dogs and cat images and groups it as two clusters – dogs and cats.

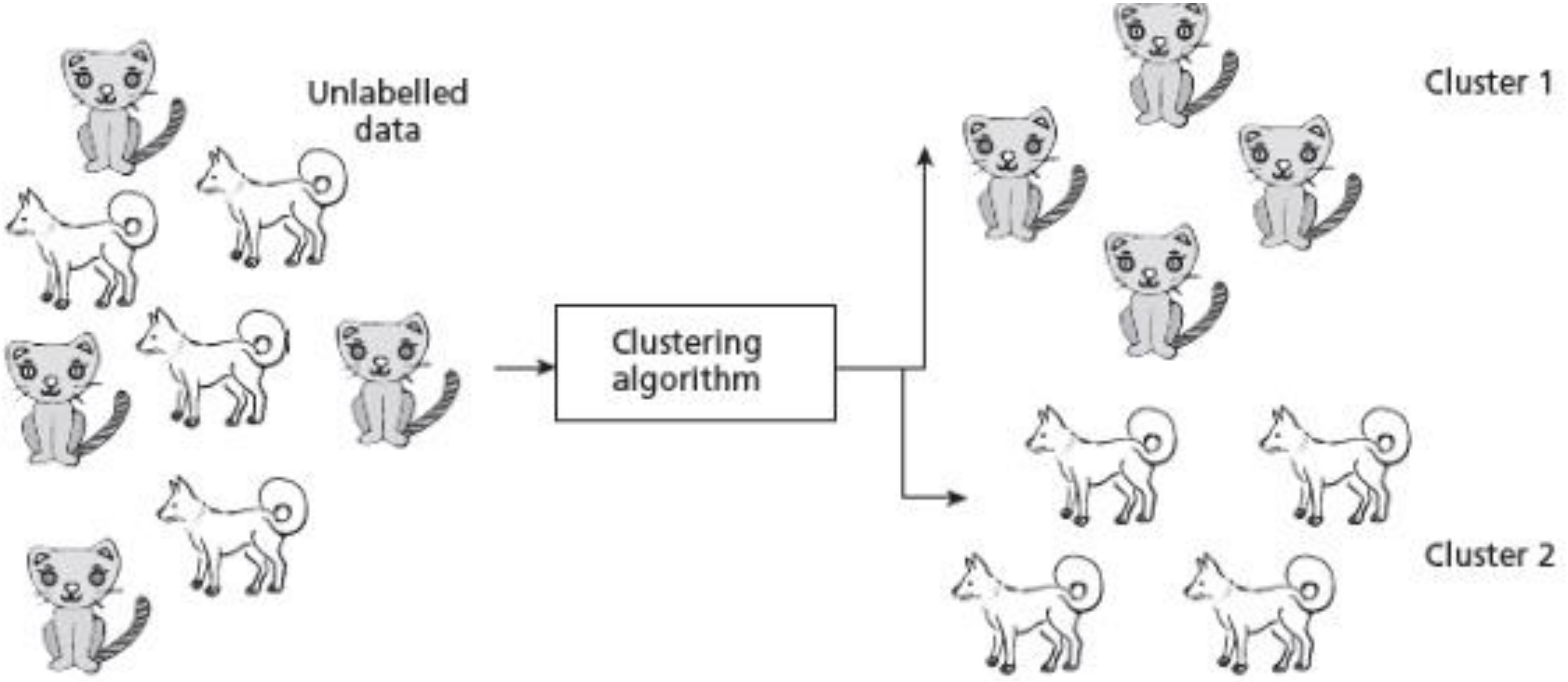


Figure 1.9: An Example Clustering Scheme

Unsupervised Learning

- **Dimensionality Reduction algorithms** are examples of unsupervised algorithms. It takes the higher dimension data as input and outputs the data in lower dimension by taking advantage of the variance of the data. It is a task of reducing the dataset with few features without losing the generality.

Semi-supervised learning

- The dataset has a huge collection of unlabeled data and some labelled data. Labelling is a costly process and difficult to perform by the humans. Semi supervised algorithms use unlabeled data by assigning a pseudo-label. Then the labelled and pseudo-labelled dataset can be combined.

Semi-supervised learning

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- Semi supervised algorithms use unlabeled data by assigning a pseudo-label.
- Then the labelled and pseudo-labelled dataset can be combined...

Reinforcement Learning

- It mimics human beings.
- Reinforcement Learning, allows the agent to interact with the environment to get rewards.
- The agent can be human, animal, robot or any independent program.
- The rewards enable the agent to gain experience.
- The agent aims to maximize the reward.
- The reward can be positive or negative.
- When the rewards are more, the behavior get reinforces and learning becomes possible.

Difference between Supervised and Unsupervised learning

Supervised Learning	Unsupervised Learning
Supervised learning algorithms are trained using labeled data.	Unsupervised learning algorithms are trained using unlabeled data.
Supervised learning model takes direct feedback to check if it is predicting correct output or not.	Unsupervised learning model does not take any feedback.
Supervised learning model predicts the output.	Unsupervised learning model finds the hidden patterns in data.
In supervised learning, input data is provided to the model along with the output.	In unsupervised learning, only input data is provided to the model.
The goal of supervised learning is to train the model so that it can predict the output when it is given new data.	The goal of unsupervised learning is to find the hidden patterns and useful insights from the unknown dataset.
Supervised learning needs supervision to train the model.	Unsupervised learning does not need any supervision to train the model.
Supervised learning can be categorized in Classification and Regression problems.	Unsupervised Learning can be classified in Clustering and Associations problems.
It includes various algorithms such as Linear Regression, Logistic Regression, Support Vector Machine, Multi-class Classification, Decision tree, Bayesian Logic, etc.	It includes various algorithms such as Clustering, KNN, and Apriori algorithm.

Challenges of Machine Learning

- Computers cannot solve **ill-posed problems**. It can deal with well-posed problems where specifications are complete and available.
- **Huge data:** Availability of quality data is a challenge. A quality data means it should be large and should not contain missing values or incorrect data.
- **High computation power:** To process the big data the computational requirement also increases. High computing units like Graphical Processing Unit(GPU) and Tensor Processing Unit(TPU) are required to execute machine learning algorithms.
- **Complexity of the algorithms:** It is challenging for ML professionals to design, select and evaluate optimal algorithms.
- **Bias/Variance:** Variance is the error of the model. This leads to the problem called bias/variance trade off. A model that fits the training data correctly but fails for the test data and lacks generalization is called overfitting. The reverse process is called underfitting, where the model fails for training data but has good generalization.

Machine Learning Process:

- Data mining solutions for business organizations is CRISP-DM(Cross Industry Standard Process- Data Mining).
- Steps involved in Machine Learning are:
- **Understanding the business:** Understanding the objectives and requirements of the business organization. Also formulates the problem for data mining process.
- **Understanding the data:** Involves data collection, study of the characteristics of the data, formulation of hypothesis, and matching of patterns to the selected hypothesis.
- **Preparation of data**: Involves producing the final dataset by cleaning the raw data and preparation of data for the data mining process. Missing values should be handled.
- **Modelling:** This step plays a role in the application of data mining algorithm for the data to obtain a model or pattern.
- **Evaluate:** Data evaluation using statistical analysis and visualization methods. The performance of the classifier is measured using the accuracy of the classifier.
- **Deployment:** Involves the deployment of results if the data mining algorithm to improve the existing process or for a new situation.

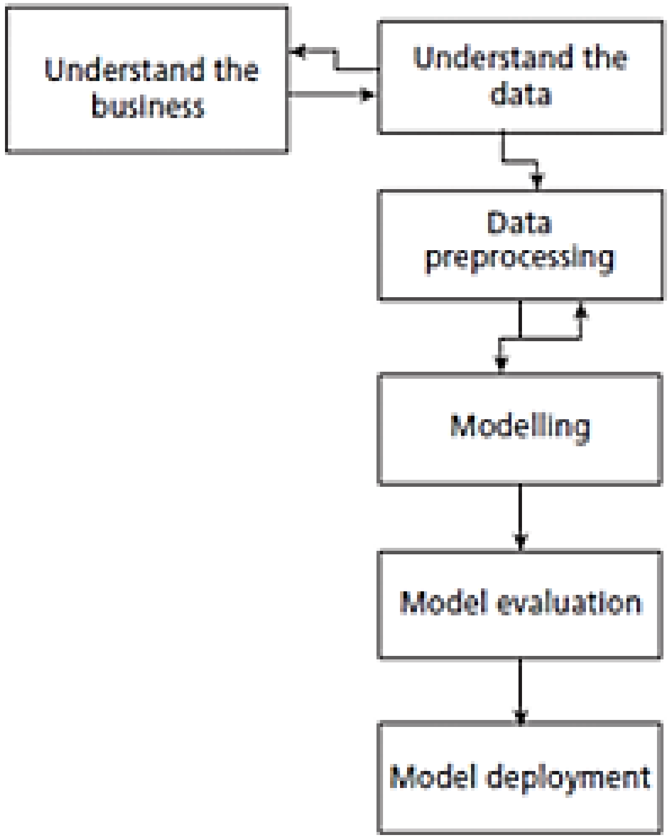


Figure 1.11: A Machine Learning/Data Mining Process

Machine Learning Applications

S.No.	Problem Domain	Applications
1.	Business	Predicting the bankruptcy of a business firm
2.	Banking	Prediction of bank loan defaulters and detecting credit card frauds
3.	Image Processing	Image search engines, object identification, image classification, and generating synthetic images
4.	Audio/Voice	Chatbots like Alexa, Microsoft Cortana. Developing chatbots for customer support, speech to text, and text to voice
5.	Telecommunication	Trend analysis and identification of bogus calls, fraudulent calls and its callers, churn analysis
6.	Marketing	Retail sales analysis, market basket analysis, product performance analysis, market segmentation analysis, and study of travel patterns of customers for marketing tours
7.	Games	Game programs for Chess, GO, and Atari video games
8.	Natural Language Translation	Google Translate, Text summarization, and sentiment analysis
9.	Web Analysis and Services	Identification of access patterns, detection of e-mail spams, viruses, personalized web services, search engines like Google, detection of promotion of user websites, and finding loyalty of users after web page layout modification
10.	Medicine	Prediction of diseases, given disease symptoms as cancer or diabetes. Prediction of effectiveness of the treatment using patient history and Chatbots to interact with patients like IBM Watson uses machine learning technologies.
11.	Multimedia and Security	Face recognition/identification, biometric projects like identification of a person from a large image or video database, and applications involving multimedia retrieval
12.	Scientific Domain	Discovery of new galaxies, identification of groups of houses based on house type/geographical location, identification of earthquake epicenters, and identification of similar land use

Chapter 2

Understanding Data

What is data?

- Data is a raw fact.
- Data can be represented in numbers, text, images, audio and video. Business organization are accumulating vast and growing amounts of data of the order of GB, TB, EXa bytes, Petabytes and so on.
- It is available in different data sources like flat files, databases or data warehouses. Data can be operational (is the one that encountered in normal business procedures and processes. Eg: daily sales data) or non operational (data that is used for decision making).
- Data by itself is meaningless. It has to be processed to generate any information.
- Processed data is called information that includes patterns, associations or relationships among data.

Elements of Big Data: 6V’s of Big data

- 1. **Volume** – Storing large amount of data in terms of petabytes(PB), exabytes(EB).
- 2. **Velocity**- The speed of data and its increase in data colume is noted as velocity. The power of internet ans IOT devices ensures that the data is arriving at a faster rate.
- 3. **Variety** – of Big data includes: Form – forms of data(text, graph, audio, video), Function- data from different sources like human conversations, transaction records and old archive data, Source of data – Data is open/public data, social media data and multimodal data.
- 4. **Veracity** – It deals with truthfulness, believability, and confidence in data. Sources of error such as technical errors, typographical errors and human errors.
- 5. **Validity** – is the accuracy of the data for taking decisions or for any other goals that are needed by the given problem.
- 6. **Value**- It indicates the value of information that is extracted from the data and its influence on the decision that are taken based on it.

Types of Data

- Structured Data
- Unstructured Data
- Semistructured Data

Types of Data

- **Structured Data:** Data is stored in an organized manner like database. Structured data encountered in ML are
- **Record Data** – A dataset is a collection of measurements taken from a process. The measurements can be arranged in the form of a matrix. Rows in a matrix represent entities or records or instances and columns represent the attributes or features or fields. The table is filled with the observed data.
- **Data Matrix** – It is a variation of the record type because it consists of numeric attributes. The standard matrix operations can be applied on these data. The data is thought as points or vectors in the multidimensional space where every attribute is a dimension describing the object.
- **Graph data** – It involves the relationships among objects. Eg: A webpage can refer another web page and can be modeled as graph. Nodes are the webpages and the hyperlink is the edge that connects the nodes.
- **Ordered data** – Ordered data objects involve attributes that have an implicit order among them Eg:
  - **Temporal data:** Data whose attribute are associated with time. Time series data.
  - **Sequence data :** It is sequential data without timestamps. Data involves the sequence of words or letters.
  - **Spatial data:** It has attributes such as positions or areas. Eg: Maps are spatial data where the points are related by location.

Types of Data

- Unstructured Data: It includes video, image and audio. It also includes textual documents, programs and blog data. Nearly 80% of data are unstructured data.
- Semistructured Data: Partially structured and partially unstructured. Eg XML/JSONdata, RSS feeds and hierarchical data.

Data storage and representation:

- The goal of data storage management is to make data available for analysis. Flat files: Simplest, cheapest and commonly available data source. The data is stored in plain ASCII or EBCDIC format. Minor changes of data in flat files affect the results of the data mining algorithms. It is suitable only for storing small dataset. Eg: CSV, TSV. Database System: It consists of database files and database management system. Database files contain original data and metadata. DBMS aims to manage data and improve operator

Data storage and representation:

- Database System: It consists of database files and database management system. Database files contain original data and metadata. DBMS aims to manage data and improve operator performance. Relational database consists of sets of tables. The tables have rows and columns. Columns represent attributes and rows represent tuples. Different types of databases are:
- Transactional database: is a collection of transactional records. Each record is a transaction. It may have a timestamp, identifier and a set of items.
- Time series database: stores time related information like log files where data is associated with a time stamp.
- Spatial databases contain spatial information in a raster or vector format. Raster format are either bitmaps or pixel maps. Vector format can be used to store maps as basic geometrical primitives like points, lines, polygons and so forth.

Data storage and representation:

- **World Wide Web:** It provides a diverse worldwide online information source. The objective of data mining algorithms is to mine interesting patterns of information present in WWW.
- **XML(eXtensible Markup Language):** It is both human and machine interpretable data format that can be used to represent data that needs to be shared across the platforms.
- **Data Stream:** It is dynamic data, which flows in and out of the observing environment. Typical characteristics of data stream are huge volume of data, dynamic, fixed order movement, and real time constraints.
- **RSS(Really simple Syndication):** It is a format for sharing instant feeds across services
- **JSON(JavaScript Object Notation):** Useful data interchange format that is used for many machine learning algorithms.

Big Data Analytics

- The primary aim of data analysis is to assist business organization to take decisions. It takes the data and generates useful information and insights for assisting the organizations.
- Data analytics refers to the process of data collection, preprocessing and analysis. It deals with the complete cycle of data management.
- Data analysis id just a part of data analytics. It takes the historical data and does the analysis.
- Data analytics concentrates more on future and helps in prediction.

Four type of data analytics:

- Descriptive analytics: It is about describing the main features of the data. After data collection is done, descriptive analytics deals with the collected data and quantifies it.
- Diagnostic analytics: It deals with the question- why? Also known as causal analysis, as it aims to find out the cause and effect of the events. For eg: if a product is not selling, diagnostic analysis aims to find out the reason.
- Predictive analytics: It deals with the future. It deals with question- what will happen in future given this data? This involves the application of algorithms to identify the patterns to predict the future.
- Prescriptive analytics: It is about the finding the best course of action for business organizations. It goes beyond prediction and helps in decision making by giving a set of actions. It helps the organizations to plan better for the future and to mitigate the risks that are involved.

Big data analysis framework:

- Big data framework is a 4-layered architecture with the following layers:
- 1. **Data connection layer**: It has data ingestion and data connectors. Data ingestion means taking raw data and importing it into appropriate data structures. It performs the tasks of ETL(Extract, Transform and Load operations).
- 2. **Data management layer:** It performs preprocessing of data. The purpose of this layer is to allow parallel execution of queries, and read, write and data management tasks. Schemes implemented by this layer such as data-in-place, where the data is not moved at all, or constructing data repositories such as data warehouses and pull data on-demand mechanisms.
- 3. **Data analytics layer:** It has many functionalities such as statistical tests, ML algorithms to understand and construction of ML models. It also implements many model validation mechanisms too.
- 4. **Presentation layer:** It has mechanisms such as dashboards, and applications that display the results of analytical engines and ML algorithms.

Big data processing cycle

- Big data processing cycle involves data management that consists of the following steps:

1. Data collection
2. Data preprocessing
3. Applications of Machine Learning algorithm
4. Interpretation of results and visualization of ML algorithm

Data collection:

- The first task of gathering datasets are the collection of data. It is often estimated that most of the time is spent for collection of good quality data. Because it yields better result. Good data has following properties:

1. Timeliness – Data should be relevant and not stale or obsolete data.
2. Relevancy – The data should be relevant and ready for the ML algorithms. All the necessary information should be available and there should be no bias in the data.
3. Knowledge about the data – The data should be understandable and interpretable and should be self-sufficient for the required application as desired by the domain knowledge engineer.

Data collection:

- The data source can be classified as
  - open/public data – Does not have any stringent copyright rules or restrictions. Eg: Government senses data.
  - social media data - Data that is generated by various social media platforms like twitter, facebook, youtube and instagram. Data generate is in huge volume.
  - multimodal data – data that involves many modes such as text, video, audio and mixed types. Eg: Image archives, WWW. These are heterogeneous in nature.

Data Preprocessing:

- In the real world, the data may be incomplete, inaccurate, outlier data, missing values, data with inconsistent values, duplicate data.
- Data preprocessing improves the quality of the data mining techniques. The raw data must be preprocessed to give accurate results.
- The process of detection and removal of errors in data is called data cleaning.
- Data wrangling means making the data processable for ML algorithms.
- Data errors may be typographical errors, incorrect measurement and structural errors like improper data formats, omission and duplication of attributes, noise in data.

Data Preprocessing

Patient ID	Name	Age	DOB	Fever	Salary
1	AAA	21		Low	-1500
2	BBB	35		High	Yes
3	CCC	5	10/10/1980	Low	" "
4	DDD	136		High	Yes

- Salary=" "is incomplete data
- DOB of 3 people are missing, age of patient id is 5 but DOB is 10/10/1980.
- Salary cannot be less than 0. It is an instance of noisy data.
- Outliers are data that exhibit the characteristics that are different from other data and have very unusual values.
- Age of patient ID cannot be 136. Outliers may be legitimate data and sometimes are of interest to the data mining algorithms.
- These errors often come during the data collection stage.
- These must be removed so that ML algorithms yield better results as the quality of results is determined by the quality of input data.
- The removal process is called data cleaning.

Missing Data Analysis

- The primary data cleaning process is missing data analysis.
- Data cleaning routines attempt to fill up the missing values, smoothen the noise while identifying the outliers and correct the inconsistencies of the data.
- This enables data mining to avoid over fitting of the models.

Missing Data Analysis

- Procedure to solve the problem of missing data:
- **Ignore the tuple:** A tuple with missing data, especially the class label is ignored. This method is not effective when the percentage of the missing values increases.
- **Fill the missing values manually** – The domain expert can analyses the data tables and carry out the analysis and fill in the values manually. But, this is time consuming and may not be feasible for larger sets.
- A **global constant** can be used to fill in the missing attributes. The missing values may be 'Unknown' or be 'Infinity'. But, some data mining result may give spurious result by analyzing these labels.
- **Attribute value** may be filled by the attribute value. Say, the average income can replace a missing value.
- Use the **attribute mean** for all samples belonging to the same class. Here the average value replaces the missing values of all tuples that fall in this group.
- Use the most **possible value** to fill in the missing value. The most probable value can be obtained from other methods like classification and decision tree prediction.

Removal of noisy or outlier data

- Noise is a random error or variance in a measured value.
- It can be removed by using binning, which is a method where the given data values are sorted and distributed into equal frequency bins.
- The bins are also called buckets.
- The binning method then uses the neighbor values to smooth the noisy data.
- Techniques to smooth the noisy data.
    - Smooth by means – mean of the bin removes the values in the bins.
    - Smoothing by bin medians – bin median replaces the bin values.
    - Smoothing by bin boundaries – bin value is replaced by the closest bin boundary.
- Maximum and minimum values are called bin boundaries.

Example:

- Set S={12, 14, 19, 22, 24, 26, 28, 31, 34} Applying various binning techniques and show the result.
- **Solution:** By equal-frequency bin method, the data should be distributed across bins.
- Assume bin size=3,
- Bin 1: 12, 14, 19 Bin2: 22, 24, 26   Bin3: 28, 31, 32
- Smoothing by means:

  Bin1: 15, 15, 15 Bin2: 24, 24, 24 Bin3: 30.3, 30.3, 30.3
- Smoothing by boundaries:

  Bin1: 12, 12, 19 Bin2: 22, 22, 26 Bin2: 28, 32, 32

Data Integration and Data Transformations:

- Data integration involves routines that merge data from multiple sources into a single data source. So this may lead to redundant data.
- The main goal of data integration is to detect and remove redundancies that arise from integration.
- Data transformations routines perform operations like normalization to improve the performance of the data mining algorithms.
- It is necessary to transform data so that it can be processed.
- One technique is normalization, the attribute values are scaled to fit in a range(0-1) to improve the performance of the algorithms.
- Normalization procedures are: min-max, z-score

Min-Max procedure

- Each variable V is normalized by its difference with the minimum value divided by the range to a new range 0-1.

$$Min - max = \frac{V-min}{max-min} * (new\ max - new\ \mathrm{min}) + new\ min$$

- Min and max are the maximum and minimum of the given data and new min are the minimum and maximum of the target range 0 and 1.

Example: V={88,90,92,94} Apply min-max procedure to map the marks to new range 0-1.

Minimum=88 maximum=94. New min=0 new max=1.

For marks 88, $\min max = \frac{88-88}{94-88} * (1-0) + 0 = 0$

For marks 90, $\min max = \frac{90-88}{94-88} * (1-0) + 0 = 0.33$

For marks 92, $\min max = \frac{92-88}{94-88} * (1-0) + 0 = 0.66$

For marks 94, $\min max = \frac{94-88}{94-88} * (1-0) + 0 = 1$

For marks{88, 90, 92, 94} are mapped to the new range {0, 0.33, 0.66, 1}

z-Score

- This procedure works by taking the difference between the field value and mean value, and by scaling this difference by standard deviation of the attribute.

$$V* = V - \mu/\sigma$$

Here, σ is the standard deviation of the list V and μ is the mean of the list V.

- Eg: V={10, 20, 30} Convert the marks to z-score Mean=20 and standard deviation=10

$$\text{z-score of } 10 = \frac{10-20}{10} = -\frac{10}{10} = -1$$

$$\text{z-score of } 20 = \frac{20-20}{10} = \frac{0}{10} = 0$$

$$\text{z-score of } 30 = \frac{30-20}{10} = \frac{10}{10} = 1$$

Hence, the z-score of the marks 10, 20, 30 are –1, 0 and 1, respectively.

Z-score

- z-scores are used to detect outlier detection.
- If the data value z-score function is either less than -3 or greater than +3, then it is possibly an outlier.
- The major disadvantage of z-score function is that it is extremely sensitive to outliers as it is dependent on mean.

Dataset and Data types:

- A dataset is collection of data objects.
- It may be records, points, vectors, patterns, events, cases, samples or observations.
- The records contain many attributes.
- An attribute can be defined as the property or characteristics of an object. The data types are shown below.
- Every attribute should be associated with a value.
- This process is called measurement.
- The type of attribute determines the data types, often referred to as measurement scale type.

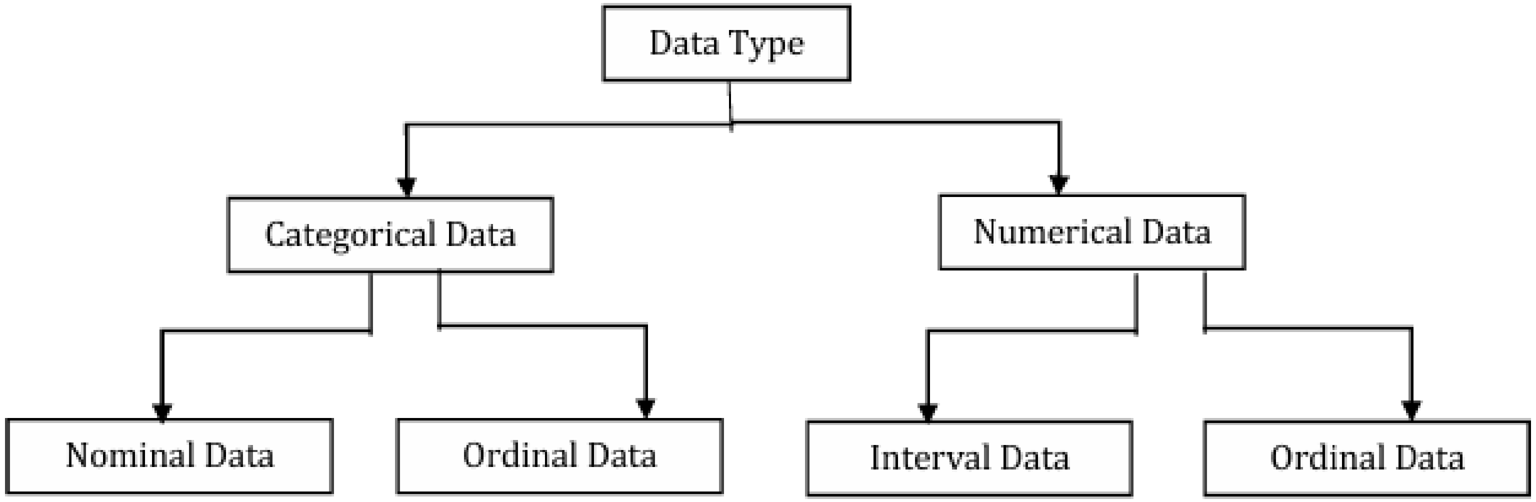


Figure: 2.1 Types of Data

Categorical or qualitative data

- Nominal Data – The patient ID is nominal data. Nominal data are symbols and cannot be processed like a number.
- For example, the average of a patient ID does not make any statistical sense.
- Nominal data type provide only information but has no ordering among data.
- Only operation like (=, ≠) are meaningful for these data.
- For example, the patient ID can be checked for equality and nothing else.
- Colors:Red, Green, Yellow
- Gender(M,F,TG), Employment Status(Employed, Unemployed, Student, Retired)
- Blood Type(A, B, O, AB), Geographical Locations

Categorical or qualitative data

- Ordinal Data – It provides enough information and has natural order.
- For example, Fever= {Low, Medium, High} is an ordinal data.
- Certainly, low is less than medium and medium is less than high, irrespective of the value.
- Any transformation can be applied to these data to get a new value.
- Eg: Customer Satisfaction-Very Dissatisfied, Dissatisfied, Neutral, Satisfied, very satisfied
- Education Level: Primary, high school, Bachelor’s, Master’s, PhD.
- Income: Low income, middle income, high income

Numerical or Quantitative Data

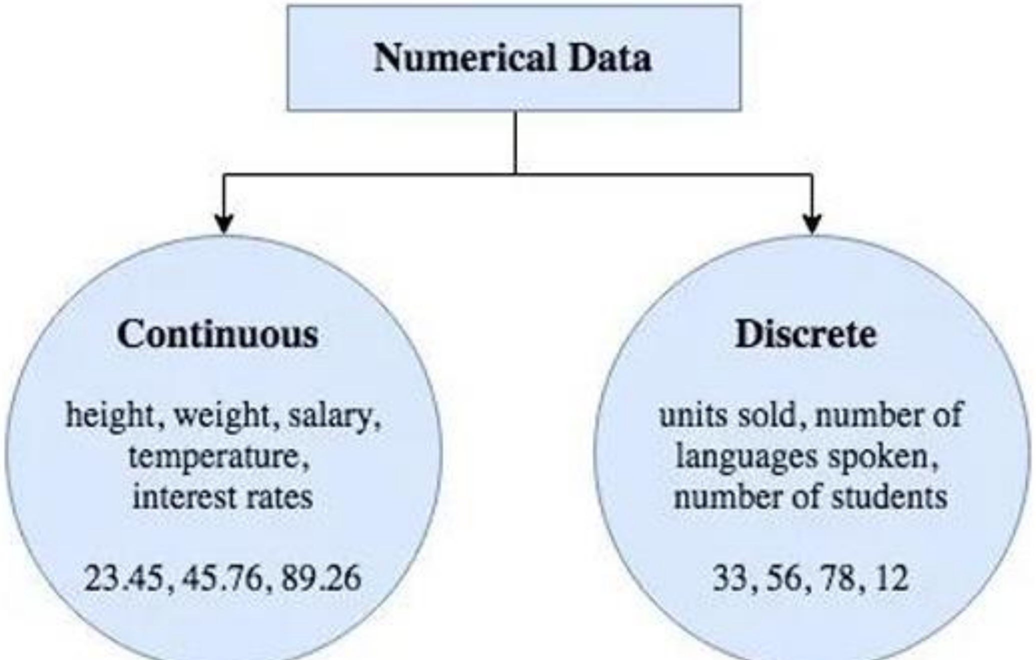
- Interval Data - Interval Data is a numeric data for which the difference between values are meaningful.
- There is no “True zero”. Zero is just an arbitrary point on the scale.
- It doesn’t mean a total absence of the thing you are measuring.
- For example there is difference between 30 degree and 40 degree. Only the permissible operation are +and -.
- Temperature: 0^{o}C doesn’t mean there is no temperature.
- Year – 0 doesn’t mean time didn’t exist

Numerical or Quantitative Data

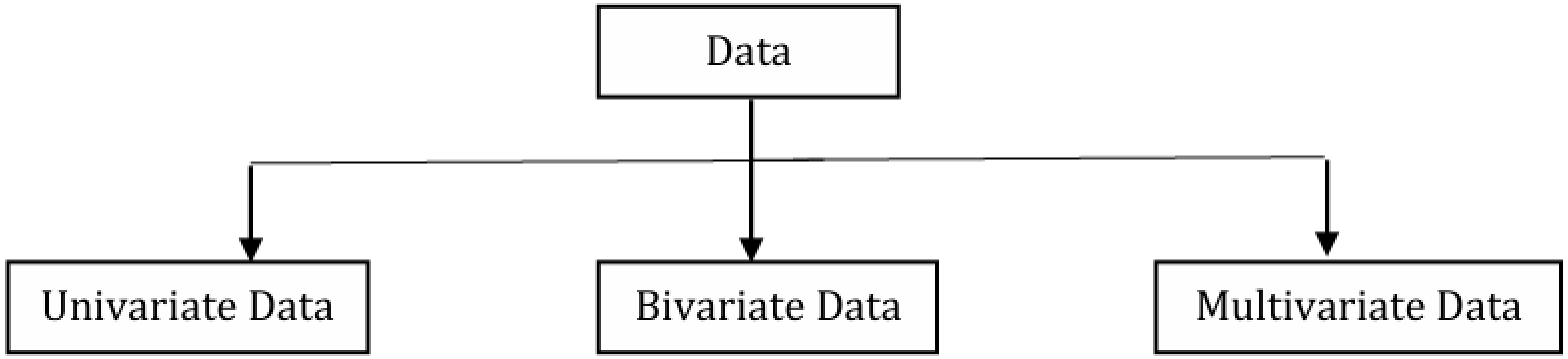
- Ratio Data – For ratio data, both difference and ratio are meaningful.
- difference between the ratio and interval data is the position of zero in the scale.
- For example, take the centigrade – Fahrenheit conversion.
- The zeroes of both scales do not match. Hence, these interval data.
- Weight -0kg, Height-0cm
- Age- being 40 years old is exactly twice as old as being 20.
- Income -0 Distance-0Km.

Another way to classify data

- 1. Discrete value data – This kind of data is recorded as integers. Example- responses of some survey can be discrete.
- 2. Continuous data – It can be fitted into a range and includes decimal point. Age, height, weight are examples for continuous data.



Based on the number of variables used in the dataset.



Univariate Data analysis and Visualization

- Univariate analysis is the simplest form of statistical analysis.
- Dataset has only one variable.
- It does not deal with cause or relationships.
- The aim of univariate analysis is to describe data and find patterns.
- Univariate data description involves finding the frequency distributions, central tendency measures, dispersion or variation, and shape of the data.

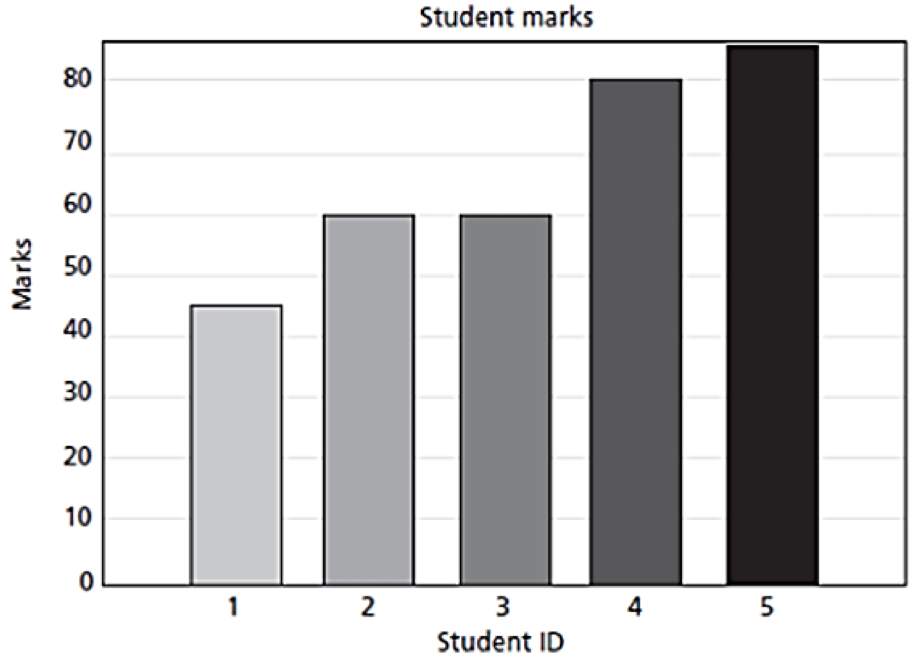
Data visualization

- Data visualization helps to understand data.
- It helps to present information and data to customers.
- Graphs that are used in univariate data analysis are bar charts, histograms, frequency polygons and pie charts.
- Advantages of graphs are
- Presentation of data, summarization of data, description of data, exploration of data, and to make comparisons of data.

Bar Chart:

- It is used to display the frequency distribution for variables.
- It is used to illustrate discrete data.
- It can also help to explain the counts of nominal data.
- It helps in comparing the frequency of different groups.

The bar chart for students' marks {45, 60, 60, 80, 85} with Student ID = {1, 2, 3, 4, 5} is shown below in Figure 2.3.



Pie Chart

Pie Chart These are equally helpful in illustrating the univariate data. The percentage frequency distribution of students' marks {22, 22, 40, 40, 70, 70, 70, 85, 90, 90} is below in Figure 2.4.

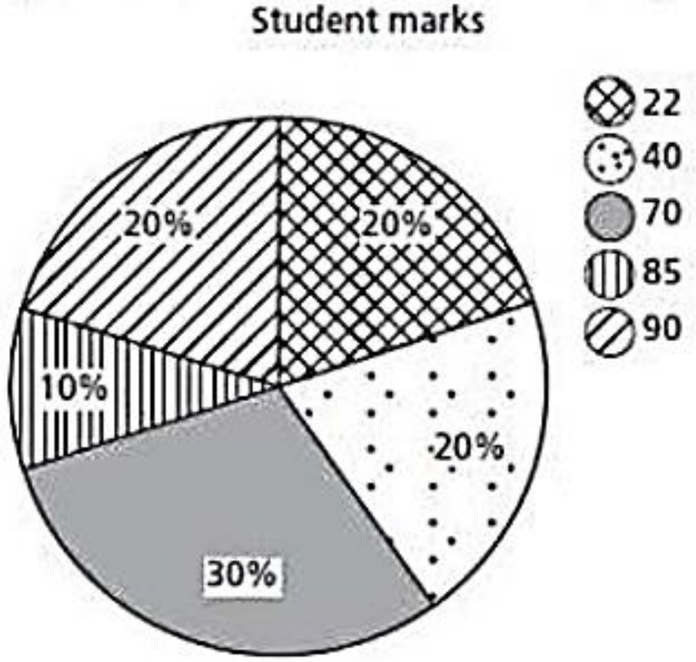


Figure 2.4: Pie Chart

It can be observed that the number of students with 22 marks are 2. The total number of students are 10. So, 2/10 × 100 = 20% space in a pie of 100% is allotted for marks 22 in Figure 2.4.

Histogram

- It plays an important role in data mining for showing frequency distributions.
- The histogram for students' marks {45, 60, 60, 80, 85} in the group range of 0-25, 26-50, 51-75, 76 100.
- Histogram conveys useful information like nature of data and it mode
- Mode indicates the peak of dataset.
- It can be used as charts to show frequency, skewness present in the data and shape.

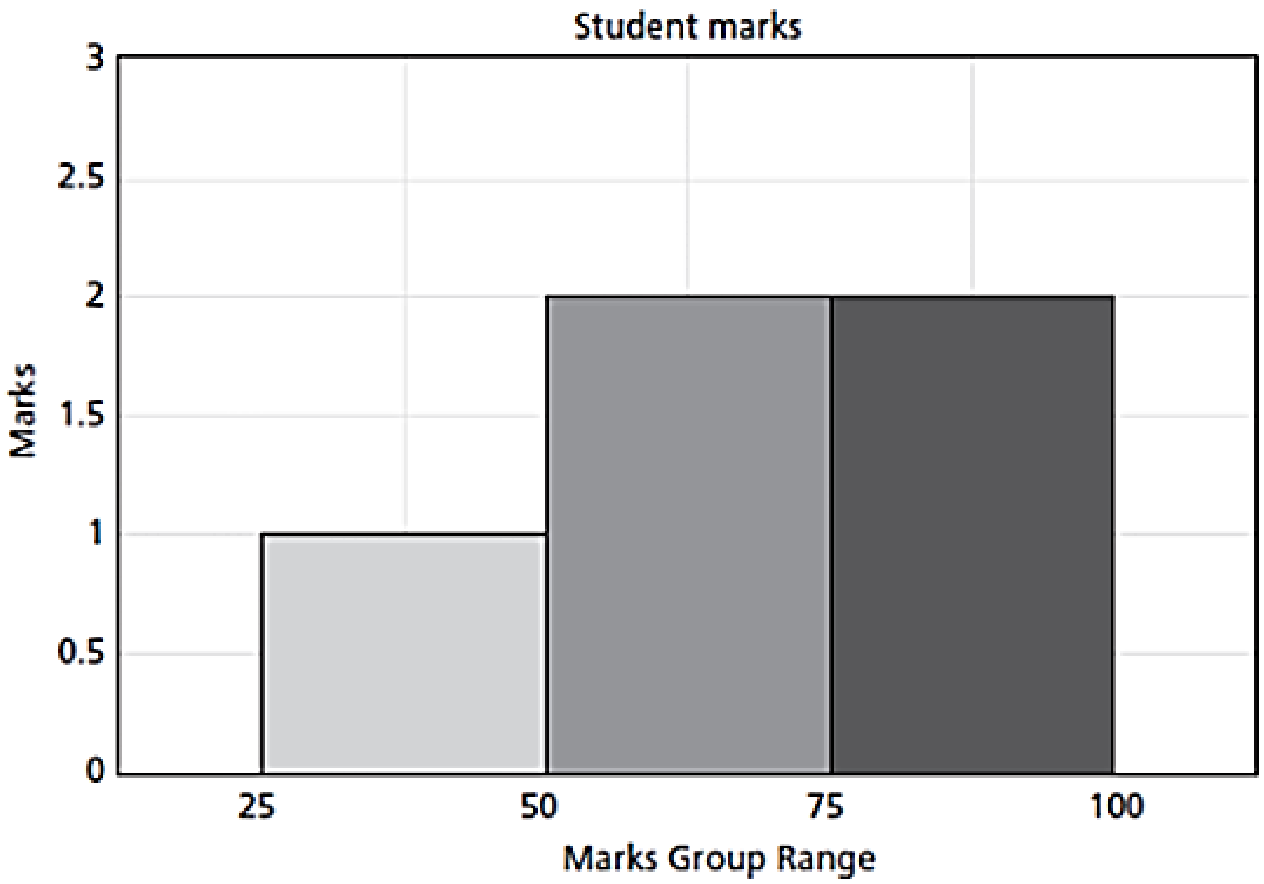
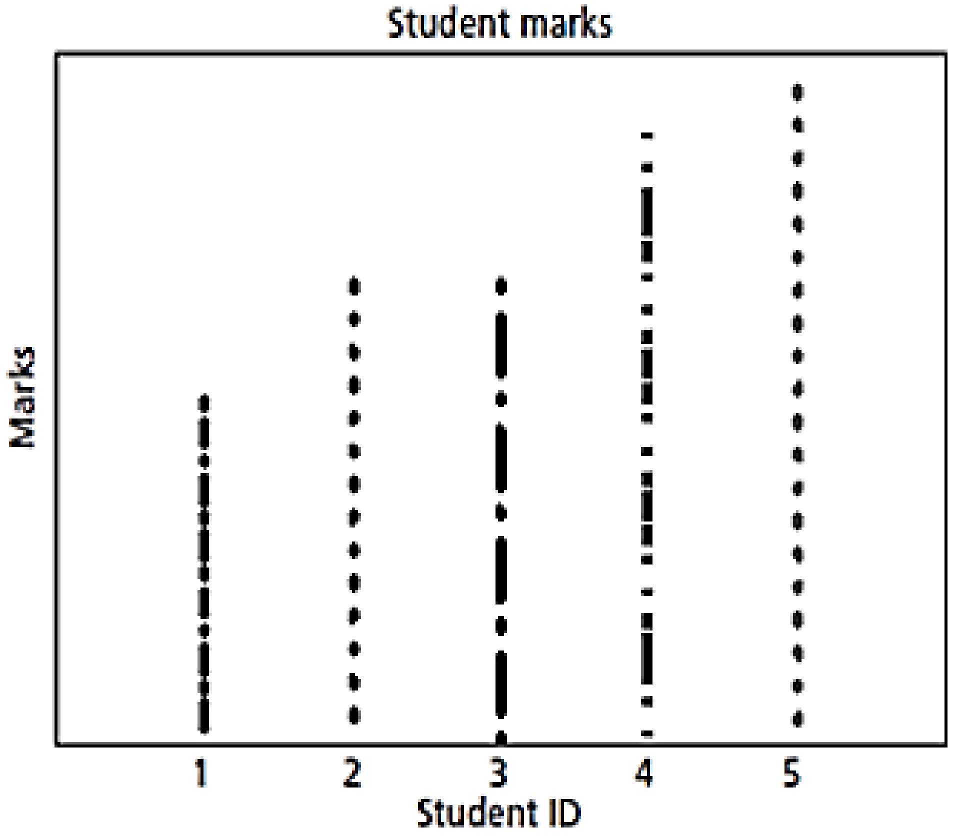


Figure 2.5: Sample Histogram of English Marks

Dot Plots

- These are similar to bar charts. They are less clustered as compared to bar charts. They are represented as single points.



Central Tendency

- Summarization of the data is called central tendency.
- Central tendency can explain the characteristics of data and helps in comparison.
- Mass data have the tendency to concentrate at certain values normally in central location. It is called measure of central tendency.
- Popular measures are mean, median, and mode.

Mean

- Arithmetic average is a measure of central tendency that represents 'center' of the dataset.
- It is calculated by adding all the data and dividing the sum by the number of observations.
- Average of all the values in the sample is denoted as $\bar{x}$.
- Let x_1, x_2, …. x_N be a set of N values or observations then arithmetic mean is given as:

$$\bar{x} = \frac{x_1 + x_2 + \cdots + x_N}{N} = \frac{1}{N}\sum_{i=1}^{N} x_i$$

Mean

- Weighted mean – Arithmetic mean that gives the weightage of all items equally. The mean is computed by adding the product of proportion and group mean. It is mostly used when the sample sizes are unequal.
- Geometric mean – Let x_1, x_2, …. x_N be a set of 'N' values or observations. Geometric mean is the Nth root of the product of N items. The formula for computing geometric mean is :

$$\text{Geometric mean} = \left(\prod_{i=1}^{n} x_i\right)^{\frac{1}{N}} = \sqrt[N]{x_1 \times x_2 \times \cdots \times x_N}$$

Median

- The middle value in the distribution is called median.
- If the total number of items in the distribution is odd, then the middle value is called median.
- If the numbers are even, then the average value of two items in the centre is the median.
- The median is the value where x_i is divided into two equal halves, with half of the values being lower than the median and half higher than the median.

Mode

- Mode is the value that occurs more frequently in the dataset.
- The value that has the highest frequency is called mode.
- Mode is only for discrete data and is not applicable for continuous data as there is no repeated values in continuous data.
- The procedure for finding the mode is to calculate the frequencies for all the values in the data, and the mode is the value with the highest frequency.

Dispersion

- The spreadout of a set of data around the central tendency(mean, mode, median) is called dispersion.
- Dispersion is represented by various ways such as range, variance, standard deviation and standard error.
- Measures of dispersion are:
- Range: Range is difference between the maximum and minimum of values of the given list of data.
- Standard deviation: Standard deviation is the average distance from the mean of the dataset to each point. The sample for sample standard deviation is given by:

$$\sigma = \sqrt{\frac{\sum_{i=1}^{N}(x_i - \overline{x})^2}{N-1}}$$

Quartiles and InterQuartile Range

- Subdivide the dataset using coordinates.
- Percentiles are about data that are less than the coordinates by some percentage of the total value.
- K^{th} percentile is the property that the k% of the data lies at or below X_i.
- For eg median is 50^{th} percentile and can be denoted by $Q_{0.50}$. The 25^{th} percentile is called first quartile Q_1, and the 75^{th} percentile is called third quartile(Q_3). Q_2 is nothing but the median of the given data.

InterQuartile Range(IQR)

- The IQR is the difference between Q_3 and Q_1.
- $IQR=Q_3-Q_1$.
- Outliers are the values falling apart atleast by the amount 1.5xIQR above the third quartile or below the first quartile. $IQR=Q_{0.75}-Q_{0.25}$.
- Example: For patients' age list{12, 14, 19, 22, 24, 26, 28, 31, 34}, find the IQR.
- Median is in the fifth position ie 24.
- The first quartile is median of the scores below the mean i.e{12, 14,19, 22}. Hence it's the median of the list below 24.
- The median is the average of the second and third values, $Q_{0.25}$ =16.5.
- The third quartile is the median of the values above the median that is {26, 28, 31, 34}.
- $Q_{0.75}$ is the average of 7^{th} and 8^{th} score that is 28+31/2=29.5.
- Hence $IQR=Q_{0.75}-Q_{0.25}$. = 29.5-16.5 =13
- Half of IQR is called semi-quartile range(SIQR) =1/2* 13 =6.5

Five-point summary and Box Plots

- The median, quartiles Q1 and Q3, and minimum and maximum written in the order <Minimum, Q1, Median, Q3, Maximum> is known as five-point summary.
- Box plots are suitable for continuous and a nominal variable.
- Box plots can be used to illustrate data distributions and summary of data.
- It is the popular way for plotting five number summaries.
- A box plot is also known as a Box and a whisker plot.

Example 2.5: Find the 5-point summary of the list {13, 11, 2, 3, 4, 8, 9}.

Solution: The minimum is 2 and the maximum is 13. The Q_1, Q_2 and Q_3 are 3, 8 and 11, respectively. Hence, 5-point summary is {2, 3, 8, 11, 13}, that is, {minimum, Q_1, median, Q_3, maximum}.

Box plots are useful for describing 5-point summary. The Box plot for the set is given in Figure 2.7.

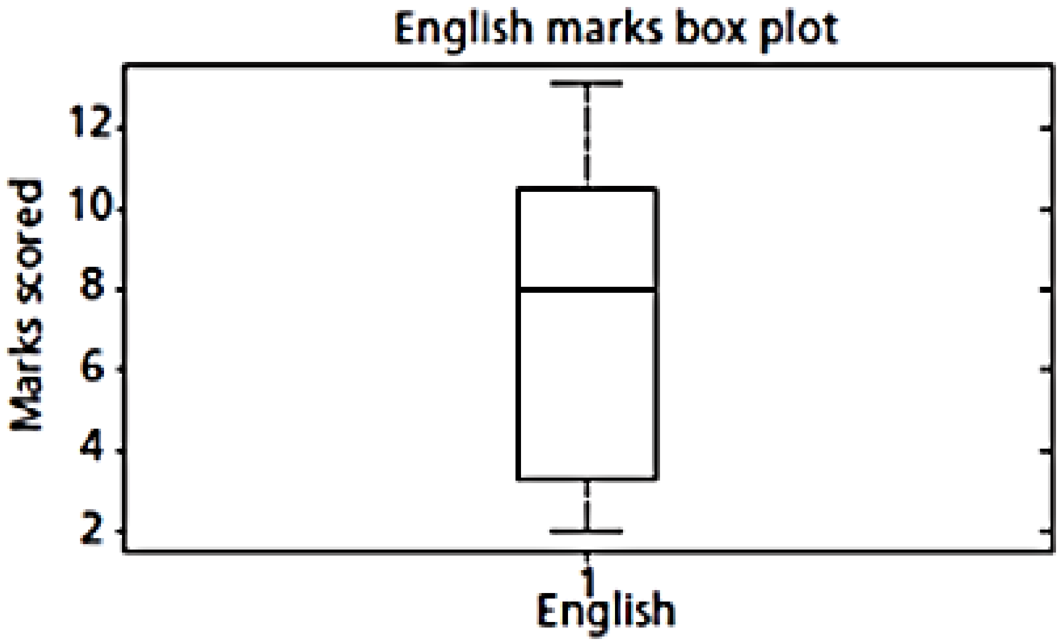


Figure 2.7: Box Plot for English Marks

Shape

- Skewness and Kurtosis(moments) indicate the symmetry/asymmetry and peak location of the dataset.
- Skewness: The measure of direction and degree of symmetry are called measures of third order.
- Ideally, skewness should be zero as in ideal normal distribution. Most of the dataset may not have perfect summary. Dataset may have very high or extremely low values.

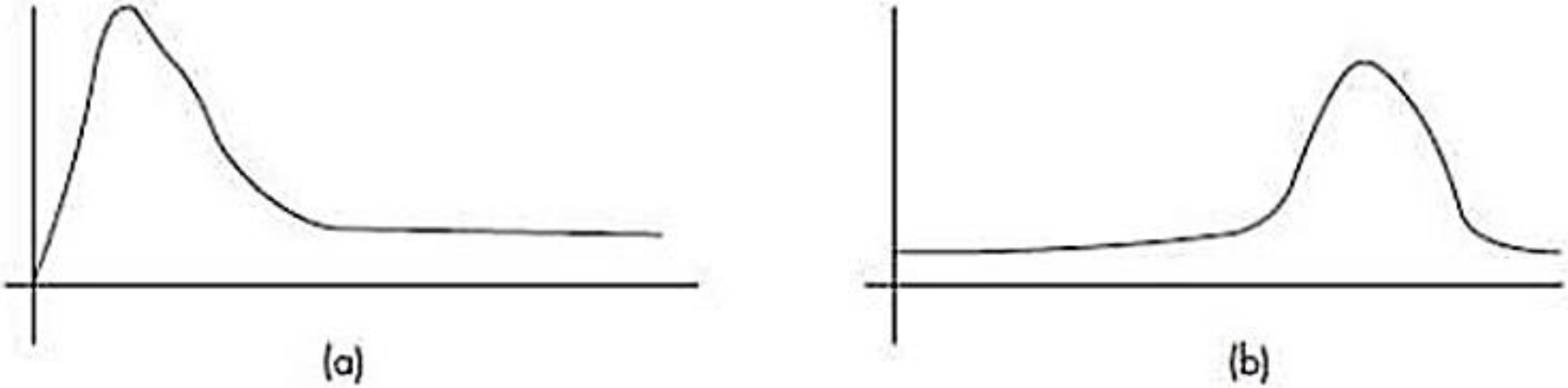


Figure 2.8: (a) Positive Skewed and (b) Negative Skewed Data

Skewness

- If the dataset has far higher values then it is said to be skewed to the right.
- If it has more low values then dataset is skewed to the left.
- If the tail is longer on the left-hand side and hump on the right hand side, it is called positive skew else it is negative skewed.
- If the data is skewed, then there is a greater chance of outliers in the dataset.
- Perfect summary means the skewness is zero.
- In the case of skew, the median is greater than the mean.
- In the positive skew, the mean is greater than the median.

Kurtosis

- It indicates the peaks of data.
- If the data is high peak, then it indicates higher kurtosis and vice versa.
- Kurtosis is the measure of whether the data is heavy tailed or light tailed relative to normal distribution.
- The normal distribution has bell-shaped curve with no long tails.
- Low kurtosis tends to have light tails implies there is no outliers.
- Kurtosis is measured using:

$$\frac{\sum_{i=1}^{N}(x_i - \bar{x})^4 / N}{\sigma^4}$$

Mean Absolute Deviation(MAD)

- MAD is another dispersion measure and is robust to outliers.
- The outlier point is detected by computing the deviation from median and by dividing it by MAD.
- The absolute deviation between the data and mean is taken.
- The absolute deviation is given as: $|x-\mu|$.
- The sum of the absolute deviations is given as $\Sigma\ |x-\mu|$.
- Mean absolute deviation is given as $\Sigma\ |x-\mu|/N$.
- Coefficient of variation(CV): Used to compare datasets with different units. CV is the ratio of standard deviation and mean.

Special Univariate plots

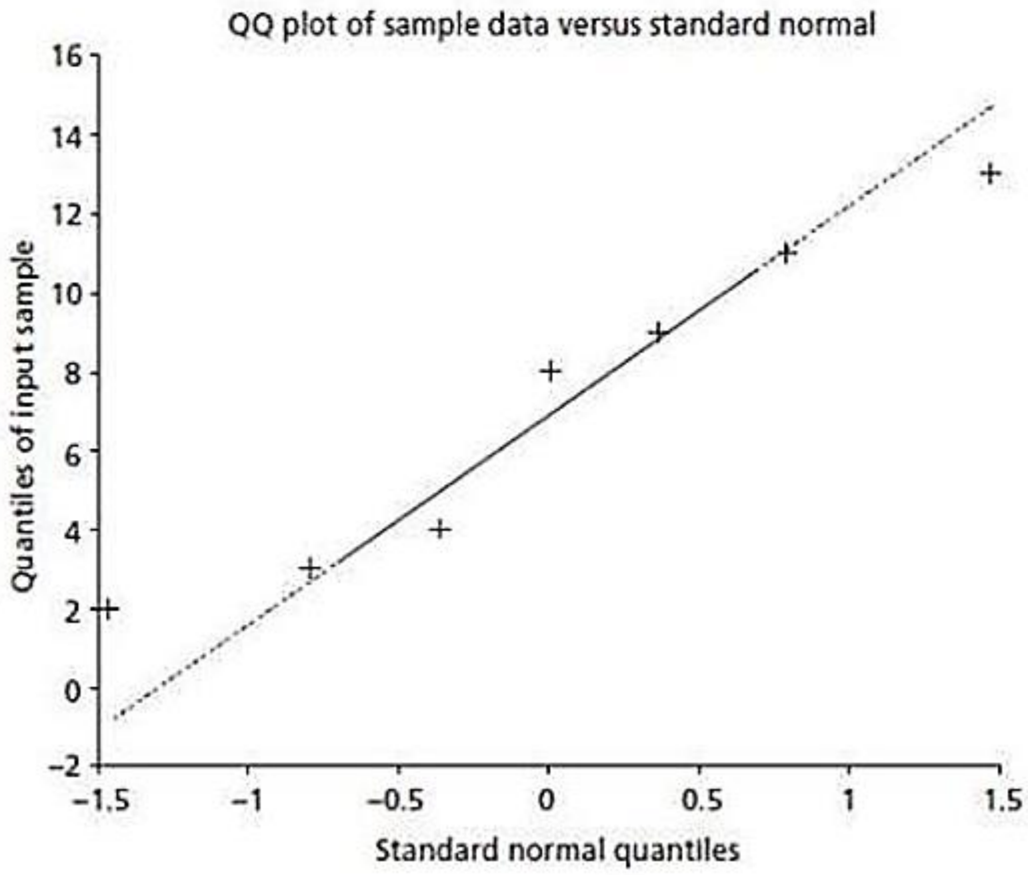
- a stem and leaf plot - to check the shape of the dataset and distribution of data.
- In this method, each value is split into a stem and a leaf. The last digit is the leaf and digits to the left of the leaf form the stem.
- Create stem leaf plots – 15, 27, 8, 17, 13, 22, 24, 25, 13, 36, 32, 32, 32, 28, 43, 7

The stem and leaf plot for the English subject marks {45, 60,60,80, 85}

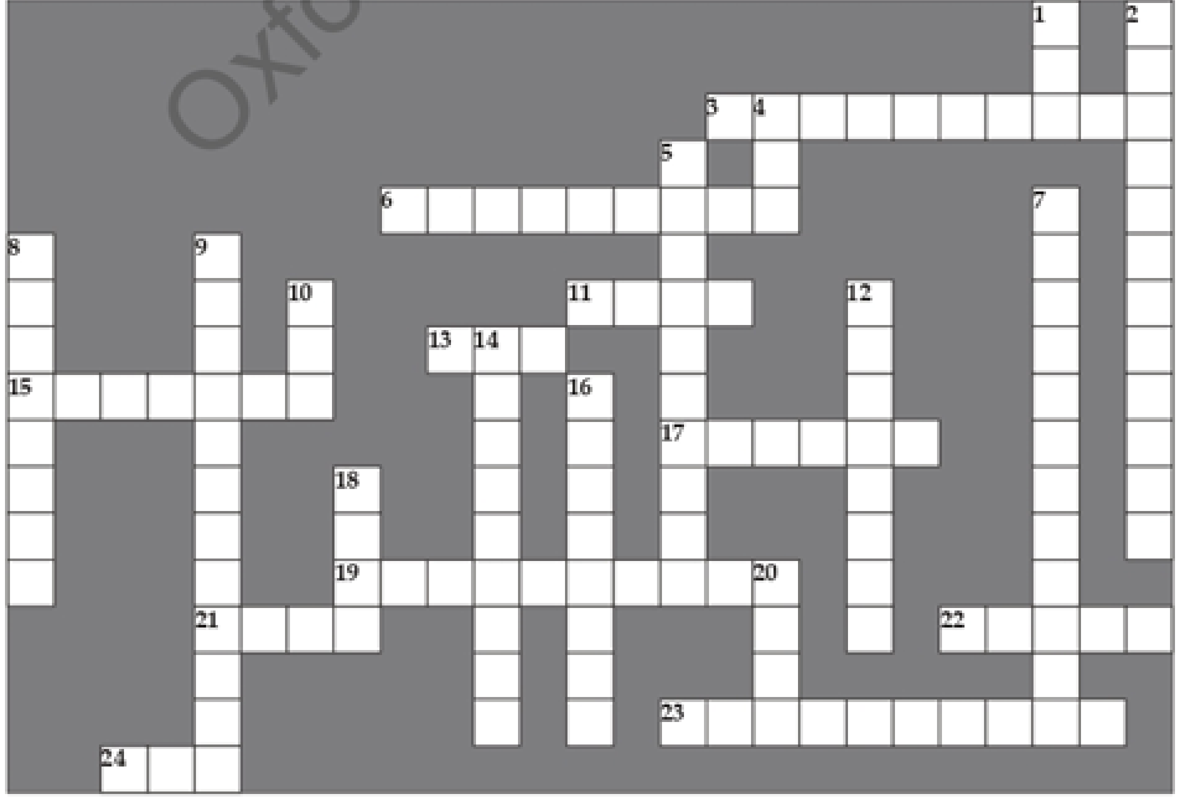
Stem	Leaf
4	5
5	
6	0 0
7	
8	05

Q-Q plots(Quantile-Quantile plots)

- Is a graphical method for determining if a dataset follows a certain probability distribution or whether two samples of data came from the same population or not.
- The Q-Q plot is a 2D scatter plot of an univariate data against theoritcal normal distribution data or of two datasets – the quartiles of the first and second datasets. The normal Q-Qplot for marks x={13 11 2 3 4 8 9}
- The points fall along the reference line if the data follows normal distribution. If the deviation is more then there is greater evidence that the datasets follow some different distribution.



1. For a given univariate dataset $S = \{5, 10, 15, 20, 25, 30\}$ of marks. Find mean, median, mode, standard deviation and variance.
2. For a given univariate dataset $S = \{5, 10, 15, 20, 25, 30\}$ of marks. Find arithmetic mean, geometric mean.



Down

1. A problem that has well-posed specification can be solved using machine learning algorithms —. (Yes/No)
2. Cluster analysis is an example of __________ learning.
4. Learning from data is the aim of statistics —. (Yes/No)
5. Predictive models can predict based on __________ data.
7. Regression can predict __________ variables.
8. Learning is __________ to the environment.
9. Bias and variance cause overfitting and __________ of model.
10. Lack of generalization in machine learning

12. Supervised learning uses __________ data.
14. Machine learning is __________ learning without being explicity programmed.
16. Model is a description of __________.
18. A semi-supervised algorithm assigns a pseudo-label for unlabelled data —. (True/False)
20. Machine learning using neural networks from a domain called artificial neural network and __________ learning.

Across

3. The initial assumption of the experiment is called a __________.
6. A study that deals with the analysis of data is called __________.
11. A domain of study that covers all the aspects of data management is called __________ science.
13. Data is a __________ fact.
15. CRISP-DM is a __________ model.
17. Pattern recognition is used for identifying patterns in __________ and videos.
19. Unsupervised learning uses __________ data.
21. Reinforcement learning uses feedback from environment for learning —. (True/False)
22. Amazon Alexa is a __________ assistant.
23. Classification is an example of __________ learning.
24. __________ data has the characteristics such as volume, variety and velocity.